# MILP Formulation for Agroforestry Biomass Supply Chain (v11)

Multi-Cycle Harvesting with Age Lag Constraints

SmartAgroforst

January 2026

## Contents



## 1 Index Sets

$$i \in \mathcal{I} = \{1, \ldots, n_s\} \quad \text{Agroforestry sites} \tag{1}$$
$$j \in \mathcal{J} = \{1, \ldots, n_j\} \quad \text{Processing facilities} \tag{2}$$
$$k \in \mathcal{K} = \{1, \ldots, n_k\} \quad \text{Consumer sites} \tag{3}$$
$$t \in \mathcal{T} = \{1, \ldots, T_{\max}\} \quad \text{Time periods} \tag{4}$$
$$t \in \mathcal{T}^{harv} = \{A^{min} + 1, \ldots, T_{\max}\} \quad \text{Harvesting periods} \tag{5}$$
$$p \in \mathcal{P} = \{1, 2, 3\} \quad \text{Product types} \tag{6}$$
$$(p, p') \in \mathcal{Q} = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\} \quad \text{Product conversion} \tag{7}$$
$$\mathcal{S}^{est} = \{(0, t) \mid t \in \{1, \ldots, T^{\max} - A^{\min}\}\} \tag{8}$$
$$\mathcal{S}^{harv} = \{(s, t) \mid s, t \in \mathcal{T} \wedge t - s \in \{A^{\min}, \ldots, A^{\max}\}\} \tag{9}$$
$$\mathcal{S}^{term} = \{(s, T^{\max} + 1) \mid s \in \{A^{\min} + 1, \ldots, T_{\max}\}\} \tag{10}$$
$$\mathcal{S} = \mathcal{S}^{est} \cup \mathcal{S}^{harv} \cup \mathcal{S}^{term} \tag{11}$$

## 2 Decision Variables

$$z_{ist} \in [0, AREA_i] \qquad \text{size of AFS of site} i \text{used in periods} t, s \text{ with } (s,t) \in \mathcal{S} \tag{12}$$

$$X_{ijpt} \geq 0 \qquad \text{Flow of product } p \in \mathcal{P} \text{ from } i \to j \text{ in } t \in \mathcal{T}^{harv} \tag{13}$$

$$X_{jkpp't} \geq 0 \qquad \text{Conversion flow of } p \text{ to } p' \text{ from } j \to k \text{ in } t \in \mathcal{T}^{harv} \tag{14}$$

$$S_{jpt} \geq 0 \qquad \text{Inventory at storage } j \text{ in period } t \in \mathcal{T} \tag{15}$$

---

## 3 Objective Function

$$\begin{aligned}
\max\ Z \ = & \sum_{k\in\mathcal{K}}\sum_{p\in\mathcal{P}} R_{kp} \cdot \sum_{j\in\mathcal{J},} \sum_{(p',p)\in\mathcal{Q}} \sum_{t\in\mathcal{T}} X_{jkp'pt} \\
& - \sum_{i\in\mathcal{I}} c_i^{est} \cdot \sum_{(0,t)\in\mathcal{S}^{est}} z_{i0t} \\
& - \sum_{i\in\mathcal{I}} (c_i^{main} + c_i^{opp}) \cdot (t-s) \cdot \sum_{(s,t)\in\mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i\in\mathcal{I}} c_i^{harv} \cdot \sum_{(s,t)\in\mathcal{S}^{harv}} z_{ist} \\
& - \sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{J}}\sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} c_p^{\text{tr-raw}} \cdot d_{ij} \cdot X_{ijpt} \\
& - \sum_{j\in\mathcal{J}}\sum_{k\in\mathcal{K}}\sum_{(p,p')\in\mathcal{Q}}\sum_{t\in\mathcal{T}} c_{p'}^{\text{tr-pre}} \cdot d_{jk} \cdot X_{jkpp't} \\
& - \sum_{j\in\mathcal{J}}\sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} c_j^{stor} \cdot S_{jpt}
\end{aligned} \tag{16}$$

---

# 4 Constraints

$$\sum_{(0,t)\in\mathcal{S}^{est}} z_{i0t} \leq AREA_i \qquad \forall\, i \in \mathcal{I} \tag{17}$$

$$\sum_{(s,t)\in\mathcal{S}} z_{ist} = \sum_{(t,u)\in\mathcal{S}} z_{itu} \qquad \forall\, i \in \mathcal{I},\ t \in \mathcal{T} \tag{18}$$

$$\sum_{j\in\mathcal{J}} X_{ijpt} \leq \sum_{(s,t)\in\mathcal{S}^{harv}} \eta_{p(t-s)} \cdot z_{ist} \qquad \forall\, i \in \mathcal{I},\ p \in \mathcal{P},\ t \in \mathcal{T}^{harv} \tag{19}$$

$$S_{jpt} = S_{jp,t-1} + \sum_{i\in\mathcal{I}} X_{ijpt} - \sum_{k\in\mathcal{K},\,(p,p')\in\mathcal{Q}} X_{jkpp't} \qquad \forall\, j \in \mathcal{J},\ p \in \mathcal{P},\ t \in \mathcal{T}^{harv} \tag{20}$$

$$\sum_{p\in\mathcal{P}} S_{jpt} \leq \mathrm{CAP}_j^{stor} \qquad \forall\, j \in \mathcal{J},\ t \in \mathcal{T} \tag{21}$$

$$\sum_{i\in\mathcal{I},\,p\in\mathcal{P}} X_{ijpt} \leq \mathrm{CAP}_j^{proc} \qquad \forall\, j \in \mathcal{J},\ t \in \mathcal{T}^{harv} \tag{22}$$

$$\sum_{j\in\mathcal{J},\,(p',p)\in\mathcal{Q}} X_{jkp'pt} \leq D_{kpt}^{\max} \qquad \forall\, k \in \mathcal{K},\ p \in \mathcal{P},\ t \in \mathcal{T}^{harv} \tag{23}$$

$$z_{ist},\ X_{ijpt},\ X_{jkp'pt},\ S_{jpt} \geq 0 \qquad \forall\, i, j, k, p, p', t \tag{24}$$